

Performance Modeling of the Ballard Mark IV Solid Polymer Electrolyte Fuel Cell: I. Mechanistic Model Development

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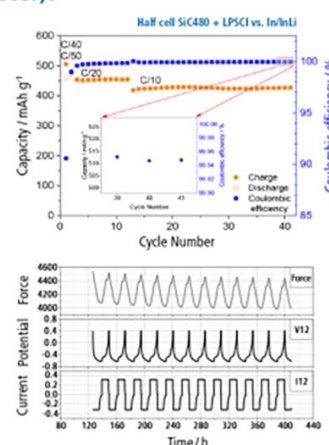
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ELECTROCHEMICAL SCIENCE AND TECHNOLOGY

Performance Modeling of the Ballard Mark IV
Solid Polymer Electrolyte Fuel Cell

I. Mechanistic Model Development

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ABSTRACT

A parametric model predicting the performance of a solid polymer electrolyte, proton exchange membrane (PEM) fuel cell has been developed using a combination of mechanistic and empirical modeling techniques. This paper details the mechanistic model development. Mass transport properties are considered in the mechanistic development via Stefan-Maxwell equations. Thermodynamic equilibrium potentials are defined using the Nernst equation. Activation overvoltages are defined via a Tafel equation, and internal resistances are defined via the Nernst-Planck equation, leading to a definition of ohmic overvoltage via an Ohm's law equation. The mechanistic model cannot adequately model fuel cell performance, since several simplifying approximations have been used in order to facilitate model development. Additionally, certain properties likely to be observed in operational fuel cells, such as thermal gradients, have not been considered. Nonetheless, the insights gained from the mechanistic assessment of fuel cell processes were found to give the resulting empirical model a firmer theoretical basis than many of the models presently available in the literature. Correlation of the empirical model to actual experimental data was very good.

A model mapping overvoltage in a proton exchange membrane (PEM) fuel cell as a function of various contributing variables would be of great utility to researchers and operators. Two common modeling approaches are: (i) theoretically based mechanistic approaches and (ii) empirically based analysis. Although, many mechanistic models can be found in the literature,¹⁻⁴ they generally require the knowledge of parameters which are not readily available, such as, transfer coefficients, humidity levels, membrane, electrode, and active catalyst layer thicknesses. Empirical models,⁵⁻⁷ on the other hand, are generally accurate over a small operating range. They fail to reflect the actual processes involved in fuel cell operation and thus are not applicable over a broad range of conditions. An approach combining both mechanistic and empirical modeling techniques has the potential to couple both mechanistic validity and the inherent simplicity of the empirical approach.

The goal of the present work was to couple a mechanistic evaluation of fuel cell processes with the application of empirical linear regression modeling techniques, in order to develop a performance model for a PEM fuel cell. A three-stage approach was followed. Initially, mechanistic modeling techniques were employed to determine the characteristic dependencies of fuel cell operation. This yielded simple algebraic expressions defining the thermodynamic equilibrium potential, as well as overvoltages due to activation, ohmic resistance, and mass transport. In the second stage, numerical parametric equations for each of the voltage loss terms were determined from experimental data. These equations reflected the form of the algebraic expressions developed in stage 1. The third stage involved the statistical analysis of the resulting parametric model.

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Stage 1 is discussed in this paper, stages 2 and 3 are discussed elsewhere⁸ and in the following paper.⁹

Physical Properties, Assumptions,
and Approximations

The fuel cell for which the present model is proposed, and with which data for the empirical analysis were gathered, is a single cell of the MkIV design produced by Ballard Power Systems Inc. of North Vancouver, BC. The cell consists of a sandwich of Du Pont's Nafion 117TM membrane between two carbon fiber paper electrodes. The faces of the electrodes contacting the membrane are impregnated with platinum catalyst and Teflon. The geometric surface area of the cell is 50.56 cm². At the time of experimentation, the cell had already logged in excess of 500 h of operation (non-continuous), and some performance deterioration had been observed in that time (output voltage reduction of approximately 0.1 V at 400 ASF for a fuel cell system operating at 70°C using 30 psig H₂ and 30 psig O₂). Over the course of experimentation (approximately 75 h), however, no further deterioration was observed, and experimental replicability was achieved. Further details on the MkIV cell can be found in various Ballard publications.¹⁰⁻¹²

The envisaged electrode structure comprises two regions; a porous, partially hydrophobic, carbon fiber-paper backing layer, and a very thin, hydrophobic layer of platinum catalyst and Teflon binder lying between the electrode and the membrane. Of this second layer, only the uppermost catalyst agglomerates (5 nm < radius < 50 nm) are believed to directly contact the membrane phase.^{1,2} It is assumed that the walls of the macropores between carbon fibers are coated with hydrophobic Teflon, while the walls of the micropores within carbon fibers will not be Teflon-coated. The wet-proofed macropores provide a path for gas flow in

the cathode, while the micropores provide a path by which liquid product water can exit to the flow channels. A similar approach to water and gas transport in the electrode has been suggested previously by Verbrugge and Bernardi.³

The hydrophobic nature of the Teflon binder is expected to cause the product water to bead. This will repel the resulting liquid microdroplets out of the catalyst layer and into the nonhydrophobic micropores of the carbon fiber-paper backing, through which they will flow to the gas flow channels. Here they will be entrained in the gas flow and removed from the fuel cell. While the Teflon is expected to be very effective in removing product water from the catalyst layer, the agglomerate theory of Ridge *et al.*² suggests that a microfilm of water will still coat the catalyst particles.

Considering the proposed physical characteristics, the expected transport processes at the cathode are: (i) diffusion of oxygen (in a mixture of O_2 , H_2O_{vap} , and possibly N_2) through the wet-proofed pores of the porous backing layer, (ii) diffusion of dissolved oxygen through a water film in the catalyst layer, and (iii) capillary and surface tension induced flow of water microdroplets through nonwet-proofed pores in the carbon fiber backing layer. It is assumed that the nonwetproofed pores are closed to gas flow at all current densities, such that the effective porosity of the cathode (in terms of gas flow) is constant. The assumption of a constant effective porosity has been employed in all of the mechanistic models reviewed thus far.¹⁻⁴

The anode and cathode gases are saturated with water vapor in a humidifier, prior to entry into the flow channels of the cell. Water from a heated reservoir is circulated through the humidifier, where both heat and humidity are transferred to the feed gases through a semipermeable membrane. The humidification water is also passed through channels in the fuel cell for heating purposes (the single cell requires heating to maintain operating temperatures in excess of 30°C) prior to returning to the reservoir. For modeling purposes, the reactant streams exiting the humidifier are assumed to be saturated with water vapor at the temperature of the humidification water exiting the fuel cell (the temperature drop of the humidification water across the fuel cell is less than 2°C), which will henceforth be termed the fuel cell operating temperature. As a simplifying approximation, isothermal properties are assumed for the flow streams in the flow channels, and it is assumed that the product water generated at the cathode will be in the liquid state and in equilibrium with the water vapor in the cathode gas stream. In actual fuel cell systems, and especially in multicell stacks, thermal gradients are likely to apply, due to the heats of reaction at the electrodes, and the product water is likely to be a mixture of liquid and vapor. With the available apparatus, however, it was not possible to characterize these gradients or the state of the product water.

It is expected that a gradient of water concentration will exist across the electrolyte membrane, with the level of hydration increasing from the anode to the cathode. This is expected to lead to diffusion of water from the cathode to the anode. A constant boundary condition of full hydration is assumed for the membrane at the cathode/membrane interface (*i.e.*, the membrane at the interface is fully wetted and will have a constant water content), due to the production of water in the cathode catalyst layer.

Due to the water concentration gradient in the membrane, the anode side of the membrane is not expected to be fully wetted during fuel cell operation. As a result some absorption of water from the humidified fuel feed is expected. This will progressively reduce the vapor pressure of water in the anode gas along the length of the anode flow channel.^{13,14} The only direct mathematical use of the anode stream water vapor pressure in the model is in the calculation of the thermodynamic equilibrium potential, where small deviations in hydrogen partial pressure are of negligible consequence. An effective anode stream water vapor pressure of 50% of the saturation vapor pressure is there-

fore assumed in all mathematical calculations as a first approximation.

Development of the Mechanistic Model

Determination of mass transport effects.—Before thermodynamic and kinetic equations can be developed, the effective concentrations of hydrogen and oxygen at the catalyst surfaces must be determined. Three distinct mass transport interfaces exist in both electrodes: (I) between the gas flow channels and the porous electrodes, (II) between the gas phase in the electrodes and any water film which may cover the catalyst sites, and (III) between the water film and the catalyst surface. To define effective reactant partial pressures or concentrations at each of these interfaces, a mass transport analysis is required.

I. Effective pressures in the flow channels.—The reactant partial pressures in the inlet flow channels will vary with both the humidification level of the inlet streams, and the consumption rates of oxygen and hydrogen. The partial pressure of water vapor in the flow channels can be defined by the saturation vapor pressure at the fuel cell operating temperature

$$p_{H_2O, cathode} = p_{H_2O}^{sat} \quad [1a]$$

$$p_{H_2O, anode} = 0.5 \cdot p_{H_2O}^{sat} \quad [1b]$$

The partial pressures of oxygen and hydrogen in the humidified inlet gas streams can be determined from the dry gas compositions and the water partial pressure

$$p_{H_2, inlet}^{hum} = \frac{(P - (p_{H_2O, anode}))}{P} p_{H_2, inlet}^{dry} \quad [2a]$$

$$p_{O_2, inlet}^{hum} = \frac{(P - (p_{H_2O, cathode}))}{P} p_{O_2, inlet}^{dry} \quad [2b]$$

The outlet dry gas composition at the cathode can be calculated as a function of the dry inlet mole fractions, the oxygen stoichiometric ratio (*SR*) (the ratio of the inlet flow rate of oxygen to the rate at which it is consumed), and the total pressure

$$p_{O_2}^{out dry} = \left[\frac{x_{O_2}^{in dry} - \frac{x_{O_2}^{in dry}}{SR}}{\left(x_{O_2}^{in dry} - \frac{x_{O_2}^{in dry}}{SR} \right) + x_{N_2}^{in dry}} \right] \cdot P \quad [3]$$

An identical calculation can be used to determine the dry gas outlet hydrogen partial pressure by substituting hydrogen and carbon dioxide mole fractions and incorporating the stoichiometric ratio for hydrogen. Equations 2a and b can then be used to determine the humidified outlet gas partial pressures for the fuel and oxidant flows by substituting outlet dry gas partial pressures for inlet dry gas partial pressures.

In a mixed flow stream (*e.g.*, O_2 , N_2 , and H_2O in the cathode or H_2 , CO_2 , and H_2O in the anode), consumption of oxygen or hydrogen at the electrode will cause concentration gradients to develop along the length of the flow channels. The shape of these concentration profiles will depend on various factors, such as reaction kinetics, flow channel geometry, flow distribution, and flow velocity. If first-order kinetics are assumed at the electrodes,¹⁵ the oxygen and hydrogen partial pressures will decline exponentially with respect to the residence time τ

$$p_{O_2} = p_{O_2}^0 e^{-k\tau} \quad [4a]$$

$$p_{H_2} = p_{H_2}^0 e^{-k\tau} \quad [4b]$$

If plug flow behavior is assumed as a first approximation, the exponential decay will depend on the velocity of gas flow in the flow channels. On the cathode side, where the consumed oxygen represents a small fraction of the total flow (stoichiometric ratios are generally >1.75 using air, *i.e.*, 733% excess flow due to the nitrogen present), velocity will be approximately constant. The effective oxygen partial pressure can then be approximated using a log-mean average of the inlet and outlet oxygen partial pressures

$$p_{O_2}^{avg} = \frac{p_{O_2,in}^{hum} - p_{O_2,out}^{hum}}{\ln \left(\frac{p_{O_2,in}^{hum}}{p_{O_2,out}^{hum}} \right)} \quad [5]$$

In the anode flow channels, on the other hand, the decline in flow velocity due to consumption and absorption is a much more significant fraction of the total velocity, since the excess flow of fuel is typically quite small ($SR = 1.15$ to 1.3). Consequently, τ will increase along the length of the flow channels and the consumption profile, Eq. 4a, will be less curved. For this model, an arithmetic mean is more likely, then, to suffice as a good first approximation of the effective hydrogen partial pressure

$$p_{H_2}^{avg} = \frac{p_{H_2,in}^{hum} + p_{H_2,out}^{hum}}{2} \quad [6]$$

If either of the dry inlet gases is undiluted (*i.e.*, no N_2 or CO_2 in the inlet stream), no average is required as the inlet and outlet partial pressures will be equal.

II. Diffusion in the electrodes.—When the effective reactant partial pressures in the flow channels are specified, diffusion of the multicomponent gas streams through the porous carbon electrodes can be described using Stefan-Maxwell equations

$$\nabla x_i = \sum_{j=1}^n \frac{RT}{PD_{ij}^{eff}} (x_i N_{j,g} - x_j N_{i,g}) \quad i = 1, 2, \dots, n \quad [7]$$

where D_{ij}^{eff} is an effective binary diffusivity for the i - j pair in the porous medium. For the purposes of the present model, only one-dimensional diffusion normal to the face of the electrodes is considered.

Two separate cases must be considered at each electrode. In the first, the inlet flows contain an inert diluent (*i.e.*, nitrogen in the cathode flow or carbon dioxide in the anode flow). In the second case the flows contain no diluent.

On the cathode side, the inlet stream in the first instance will be a mixture of oxygen and nitrogen, saturated with water vapor at the fuel cell operating temperature, *i.e.*

$$x_{O_2} + x_{N_2} + x_{H_2O,vap} = 1 \quad [8]$$

or

$$p_{O_2} + p_{N_2} + p_{H_2O,vap} = P$$

The derivative of Eq. 8 gives the relationship along the axis of oxygen diffusion into the electrode (*i.e.*, along the gas diffusion path through the electrode), *i.e.*

$$\frac{dx_{O_2}}{dz} + \frac{dx_{N_2}}{dz} + \frac{dx_{H_2O,vap}}{dz} = 0 \quad [9]$$

Assuming constant pressure in the flow channels and assuming that constant temperature maintains the partial pressure of water vapor constant at the saturation vapor pressure

$$\frac{dx_{H_2O,vap}}{dz} = 0 \quad [10]$$

Therefore the gas-phase flux of water normal to the cathode face can be set to zero

$$N_{H_2O,g} = 0 \quad [11]$$

As nitrogen is inert and cannot be removed from the diffusion channel, the net nitrogen gas-phase flux normal to the cathode face is also zero

$$N_{N_2,g} = 0 \quad [12]$$

The one-dimensional Stefan-Maxwell equation for the nitrogen component, as derived from Eq. 7, can be simplified using assumptions in Eq. 11 to 12 to give

$$\frac{dx_{N_2}}{dz} = \frac{RT}{PD_{N_2,O_2}^{eff}} (x_{N_2} N_{O_2,g}) \quad [13]$$

The gas-phase flux of oxygen can be related to current density by

$$N_{O_2,g} = \frac{I}{4F} \quad [14]$$

Combining Eq. 13 and 14 and integrating yields

$$x_{N_2}^{interface} = x_{N_2}^{channel} \cdot \exp \left[\frac{R \cdot T \cdot I \cdot \ell_d}{4 \cdot F \cdot P \cdot D_{N_2,O_2}^{eff}} \right] \quad [15]$$

where ℓ_d is the thickness of the gas diffuser portion of the electrode (~ 0.25 mm), $x_{N_2}^{channel}$ is the log mean average mole fraction of nitrogen in the humidified gas at the interface of the cathode and the cathode gas flow channel, and $x_{N_2}^{interface}$ is the effective mole fraction of nitrogen at the gas/liquid film interface of the cathode catalyst layer. The pressure diffusivity product has been estimated by Springer *et al.*,⁴ using the Slattery and Bird correlation¹⁶

$$PD_{N_2,O_2}^{eff} = 0.0002745 \left(\frac{T}{\sqrt{T_{O_2}^{crit} T_{N_2}^{crit}}} \right)^{1.832} (p_{O_2}^{crit} p_{N_2}^{crit})^{1/3} (T_{N_2}^{crit} T_{O_2}^{crit})^{5/12} \left(\frac{1}{M_{N_2}} + \frac{1}{M_{O_2}} \right)^{1/2} \epsilon \quad [16]$$

where ϵ represents a correction factor for the diffusivity to account for the void fraction, tortuosity, and varying pore cross sections of the electrode. Values for ϵ of between 0.25 and 0.35 are commonly observed in the literature for carbon fiber-paper electrodes.¹⁻³ For this model ϵ has been set to 0.3 as a simplifying approximation.

The effective partial pressure of oxygen at the gas/liquid interface of the catalyst layer can now be calculated using Eq. 8

$$p_{O_2}^{interface} = P \cdot [1 - x_{H_2O}^{sat} - x_{N_2}^{interface}] \quad [17]$$

Combining Eq. 15, 16, and 17 with appropriate values for the known constants yields⁸

$$p_{O_2}^{interface} = P \cdot \left[1 - x_{H_2O}^{sat} - x_{N_2}^{channel} \cdot \exp \left(\frac{0.291 \cdot I}{T^{0.832}} \right) \right] \quad [18]$$

A similar development for the interfacial hydrogen partial pressure on the anode side yields⁸

$$p_{H_2}^{interface} = P \cdot \left[1 - 0.5 \cdot x_{H_2O}^{sat} - x_{CO_2}^{channel} \cdot \exp \left(\frac{0.183 \cdot I}{T^{0.832}} \right) \right] \quad [19]$$

In the second case, the cathode feed stream contains only water vapor and oxygen. Now Eq. 8 indicates that the drop in oxygen partial pressure along the diffusion path will result in a corresponding drop in the total pressure. In addition, Eq. 9 indicates that the gradient in water vapor mole fraction is no longer zero (as in Eq. 10). There will therefore be a molecular flux of water vapor "out" just sufficient to balance the bulk motion of the gas due to the diffusion of oxygen "in." The net flux of water vapor through the cathode is, therefore, still zero.

Again assuming that the Stefan-Maxwell equation describes the process, Eq. 7 leads to

$$\frac{dx_{O_2}}{dz} = \frac{R \cdot T}{P \cdot D_{H_2O,O_2}^{eff}} \left(x_{H_2O} \cdot \frac{I}{4 \cdot F} \right) \quad [20]$$

Substitution and rearrangement of this equation yields

$$\frac{d(x_{O_2})}{(1 - x_{O_2})} = \frac{R \cdot T \cdot I}{4 \cdot F \cdot P \cdot D_{H_2O,O_2}^{eff}} dz \quad [21]$$

Integration of this expression from the flow channel to the catalyst surface, again taking the product $PD_{i,j}^{eff}$ as a constant, gives

$$\ln \left[\frac{(1 - x_{O_2}^{interface})}{(1 - x_{O_2}^{channel})} \right] = \frac{R \cdot T \cdot I \cdot \ell_d}{4 \cdot F \cdot P \cdot D_{H_2O,O_2}^{eff}} \quad [22]$$

Since $x_{O_2} + x_{H_2O} = 1$

$$x_{H_2O}^{interface} = x_{H_2O}^{channel} \cdot \exp \left[\frac{R \cdot T \cdot I \cdot \ell_d}{4 \cdot F \cdot P \cdot D_{H_2O,O_2}^{eff}} \right] \quad [23]$$

As the partial pressure of water vapor in the cathode is fixed at the saturation level, the partial pressure of oxygen at the interface is

$$p_{O_2}^{interface} = (1 - x_{H_2O}^{interface}) \cdot \left(\frac{p_{H_2O}^{sat}}{x_{H_2O}^{interface}} \right) \quad [24]$$

Combining Eq. 23 and 24 yields

$$p_{\text{O}_2}^{\text{interface}} = (p_{\text{H}_2\text{O}}^{\text{sat}}) \cdot \left[\frac{1}{\exp\left(\frac{R \cdot T \cdot I \cdot \ell_d}{4 \cdot F \cdot P \cdot D_{\text{H}_2\text{O},\text{O}_2}^{\text{eff}}\right) \cdot x_{\text{H}_2\text{O}}^{\text{channel}}} - 1 \right] \quad [25]$$

Evaluation of the Slattery-Bird equation for the $PD_{\text{H}_2\text{O},\text{O}_2}^{\text{eff}}$ constant, using the appropriate critical values and constants for a water/oxygen mixture finally yields⁸

$$p_{\text{O}_2}^{\text{interface}} = (p_{\text{H}_2\text{O}}^{\text{sat}}) \cdot \left[\frac{1}{\exp\left(\frac{(4.192 \cdot I)}{T^{1.334}}\right) \cdot x_{\text{H}_2\text{O}}^{\text{channel}}} - 1 \right] \quad [26]$$

A similar development for hydrogen on the anode side yields the equation⁸

$$p_{\text{H}_2}^{\text{interface}} = (0.5 \cdot p_{\text{H}_2\text{O}}^{\text{sat}}) \cdot \left[\frac{1}{\exp\left(\frac{(1.653 \cdot I)}{T^{1.334}}\right) \cdot x_{\text{H}_2\text{O}}^{\text{channel}}} - 1 \right] \quad [27]$$

III. Diffusion through a water film.—It is necessary to analyze the transport of dissolved gases through any water film which may cover catalyst sites in the anode or cathode catalyst layer. A required first step is the determination of the likely thickness of such a film, and its effect on mass transport rates. It is first assumed that any water layer on the anode side will be extremely thin, as no water is being produced at the anode, and the electrolyte membrane is very retentive of any water it contains. This film is not expected to present a significant barrier to hydrogen diffusion to the catalyst, and it can be assumed that any dissolved hydrogen gas at the anode will be in equilibrium with the bulk gas. The effective concentration can be defined by a Henry's law equation of the form

$$c_{\text{H}_2}^* = \frac{p_{\text{H}_2}^{\text{interface}}}{1.09 \cdot 10^6 \exp\left(\frac{77}{T}\right)} \quad [28]$$

The Henry's law constant in the right denominator of Eq. 28 has been determined from available solubility data.¹⁷

Contrary to the anode case, the production of water at the cathode makes the existence of a significantly thick water layer much more probable. However, data for a MkIV fuel cell using a Dow experimental membrane indicates that current densities in excess of 6000 ASF can be achieved with pure hydrogen and oxygen feeds at 3.83 atm abs and 358 K.¹⁰ Even at 6000 ASF there is no indication of a diffusion limit to the current. Thus the limiting cathodic current density must be greater than 6000 ASF. Diffusion of oxygen through a thin water film would be defined by Fick's law

$$N_{\text{O}_2} = -D_{\text{O}_2} \frac{dc_{\text{O}_2}}{dz} \quad [29]$$

where the concentration of dissolved oxygen at the gas/liquid interface can be defined by a Henry's law expression of the form

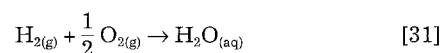
$$c_{\text{O}_2}^{\text{interface}} = \frac{p_{\text{O}_2}^{\text{interface}}}{5.08 \cdot 10^6 \exp\left(\frac{-498}{T}\right)} \quad [30]$$

The Henry's law constant in the denominator is again derived from readily available solubility data.¹⁷ The diffusivity of oxygen in water at 358 K is approximately $8 \times 10^{-5} \text{ cm}^2/\text{s}$.¹⁷ For an undiluted oxygen/water inlet stream at 358 K and 3.83 atm abs., with a current density of 6000 ASF ($6.46 \text{ A}/\text{cm}^2$), Eq. 26 defines the interfacial oxygen pressure to be 2.15 atm abs. Thus Eq. 30 defines the interfacial oxygen concentration as $1.7 \times 10^{-6} \text{ mol}/\text{cm}^3$. Equation 14 defines the oxygen flux for a current density of $6.46 \text{ A}/\text{cm}^2$ as being $1.67 \times 10^{-5} \text{ mol}/\text{cm}^2 \cdot \text{s}$. The one-dimensional concentration gradient in the water film normal to the cathode face⁴ is calculated for Eq. 29 as $0.21 \text{ mol}/\text{cm}$. Thus if the

water film thickness was $8.1 \times 10^{-6} \text{ cm}$, the concentration of dissolved oxygen at the catalyst surface would be zero. Since existing data shows that significant voltages ($\sim 0.37 \text{ V}$) can be achieved at 6000 ASF (using the Dow membrane),¹⁰ the water film thickness can be no greater than $8.1 \times 10^{-6} \text{ cm}$ and is likely to be significantly less. The maximum current available using a Nafion membrane is limited to approximately 1200 ASF by the internal resistance of the membrane. For a given set of operating conditions, if it is assumed that the water film thickness varies linearly with current density for a given electrode, then the maximum film thickness which would be seen in a Nafion cell would be on the order of 10^{-7} cm . A film of this thickness will not present a significant barrier to oxygen diffusion to the catalyst sites. Thus the surface oxygen concentration $c_{\text{O}_2}^*$ can be approximated using the interfacial concentration $c_{\text{O}_2}^{\text{int}}$ developed in Eq. 30.

The Thermodynamic Equilibrium Potential

The overall reaction in a hydrogen/oxygen fuel cell can be written as¹⁸



The thermodynamic potential of this reaction is defined by the Nernst equation^{3,18}

$$E = E^\circ - \frac{RT}{nF} \ln [p_{\text{H}_2}^* (p_{\text{O}_2}^*)^{0.5}] \quad [32]$$

where E° represents a reference potential at unit activity, and the partial pressure terms correspond to the concentrations $c_{\text{H}_2}^*$ and $c_{\text{O}_2}^*$. The standard state defines a standard state reference potential E° (equal to 1.229 V at 298.15 K and 1 atm), and E° will vary from the standard state reference in accordance with temperature¹⁸

$$E^\circ = E_0^\circ + (T - T_0) \left(\frac{\Delta S^\circ}{nF} \right) \quad [33]$$

where T_0 is the standard state temperature (298.15 K). The entropy change of a given reaction is approximately constant (the variation in specific heat with the expected changes in temperature is minimal) and can be set to the standard state value. Thus the reference potential varies directly with temperature via¹⁸

$$E^\circ = \beta_1 + \beta_2 T \quad [34]$$

where

$$\beta_1 = 1.229 \text{ V} - \frac{298.15 \cdot \Delta S_0^\circ}{nF} \quad [35a]$$

$$\beta_2 = \frac{\Delta S_0^\circ}{nF} \quad [35b]$$

Using literature values for the standard-state entropy change, the value of β_2 in this equation can be calculated to be $-0.85 \times 10^{-3} \text{ V}/\text{K}$.¹⁸ Equation 32 can now be written as

$$E = 1.229 - 0.85 \cdot 10^{-3} (T - 298.15) + \frac{RT}{nF} \ln [p_{\text{H}_2}^* (p_{\text{O}_2}^*)^{0.5}] \quad [36]$$

Further expansion of this equation yields

$$E = 1.229 - 0.85 \cdot 10^{-3} \cdot (T - 298.15) + 4.3085 \cdot 10^{-5} \cdot T \cdot \left[\ln (p_{\text{H}_2}^*) + \frac{1}{2} \cdot \ln (p_{\text{O}_2}^*) \right] \quad [37]$$

Fuel Cell Voltage Losses

The combined effect of thermodynamics, mass transport, kinetics, and ohmic resistance determines the output voltage of the cell, as defined by

$$V = E + \eta_{\text{act}} + \eta_{\text{ohmic}} \quad [38]$$

where η represents an overvoltage, and act is short for activation. The effects of mass transport will be seen in all three terms on the right side of this equation.

Activation Overvoltage

In the solid polymer fuel cell models found in the literature to date, the kinetics of hydrogen oxidation on platinum have been considered to be very rapid.²⁻⁴ The Tafel slope [slope of overvoltage vs. $\log(i)$ plot] for the reaction at standard state (298.15 K, 1 atm) on platinum catalysts is approximately 0.03 V/decade,¹⁸ and its exchange current density is approximately 50 ASF.¹⁹ As a result, activation overvoltages remain low even at high currents. The cathode reaction, by contrast, is rather slow with a standard state exchange current density in the range of 10^{-3} ASF and a Tafel slope in excess of 0.07 V/decade.¹⁸ Thus the activation overvoltage at 298.15 K and a current density of 100 ASF would be 0.009 V for hydrogen oxidation and more than 0.35 V for oxygen reduction. Consequently, the voltage drop due to activation can be predominantly attributed to the cathode reaction.

When mass transfer effects are negligible, reaction kinetics can be described by the Butler-Volmer equation¹⁸

$$i = i_0 [e^{-\alpha n F \eta / RT} - e^{(1-\alpha) n F \eta / RT}] \quad [39]$$

Considering the general electrode process



The exchange current i_0 can be expressed as¹⁸

$$i_0 = nFAk^0 \left[\exp\left(\frac{-\Delta F_e}{RT}\right) (C_A^*)^{a(1-\alpha)} (C_B^*)^{b(\alpha)} \right] \quad [41]$$

where n is the number of electrons involved per mole of electrolyzed component in the rate controlling step, a and b represent the appropriate reaction orders of the rate controlling step with respect to the reactant and product, A is the cross-sectional surface area over which the reaction is occurring, k^0 is the intrinsic rate constant, and ΔF_e is the standard free energy of activation for the reaction.

For large values of η , the second exponential in Eq. 39 is negligible, yielding the Tafel expression

$$\eta_{act} = \frac{2.3 \cdot RT}{\alpha n F} \log \frac{i_0}{i} \quad [42]$$

Given the relatively high overvoltage of the cathode reaction, its kinetics can be described in this manner. Thus the cathodic activation overpotential can be expressed as

$$\eta_{act cathodic} = \frac{RT}{\alpha_c n F} \left[\ln \left(nFAk^0 \exp\left(\frac{-\Delta F_e}{RT}\right) (c_{O_2}^*)^{(1-\alpha_c)} (c_{H^+}^*)^{(1-\alpha_c)} (c_{H_2O}^*)^{\alpha_c} \right) - \ln(i) \right] \quad [43]$$

where α_c represents the cathodic transfer coefficient. The concentration of protons in the electrolyte membrane, $c_{H^+}^*$, will be fixed by water content and the law of electroneutrality, which requires that

$$z_m c_m + \sum_i z_i c_i = 0 \quad [44]$$

where $z_m c_m$ is the total charge on the solid membrane and $\sum_i z_i c_i$ is the total charge on mobile species in the electrolyte. Protons are the only ionic species, and sulfonic acid ions are the only charged membrane species. As z_m and z_i are both constants (-1 and $+1$, respectively)

$$c_m = c_i \quad [45]$$

As the sulfonic acid ions are fixed to the backbone of the membrane's polymer chains, the number of these ions in a given portion of dry membrane will be fixed. The dry membrane is compressible, however, such that different parts of the membrane can hold different concentrations of water, and thus different concentrations of sulfonic acid ions, while still being wet throughout. The membrane region

nearest to the cathode, with constant full hydration, will have the lowest concentration of protons, while the membrane region nearest to the anode will have the highest. Given the assumption of constant water concentration at the membrane/cathode interface, the proton concentration in Eq. 43 will be constant, as will the terms k^0 , A , n , F , R , ΔF_e , and α_c . The only remaining variables are the concentration of oxygen at the reaction sites, the current, and the temperature. Grouping constants, Eq. 43 can be expressed in parametric form as

$$\eta_{act cathodic} = \beta_1 + \beta_2 T + \beta_3 T \ln(c_{O_2}^*) + \beta_4 T \ln(i) \quad [46]$$

where the constant parametric coefficients are

$$\beta_1 = \frac{(-\Delta F_e)}{\alpha_c n F} \quad [47a]$$

$$\beta_2 = \frac{R}{\alpha_c n F} \ln [nFAk^0 c_{H^+}^{*1-\alpha_c} c_{H_2O}^{*\alpha_c}] \quad [47b]$$

$$\beta_3 = \frac{R}{\alpha_c n F} (1 - \alpha_c) \quad [47c]$$

$$\beta_4 = -\frac{R}{\alpha_c n F} \quad [47d]$$

The activation overvoltage of the hydrogen oxidation reaction has been defined as¹⁸

$$\eta_{act anodic} = \frac{RT}{2F} \ln \left(\frac{i_0}{i} \right) \quad [48]$$

As the rate controlling step at the anode is likely to be chemisorption of hydrogen, Berger¹⁸ suggests that the exchange current can be defined as

$$i_{0, anodic} = 4FAk^0 \exp\left(\frac{-(\Delta F_{ec} - 0.5\Delta F_c)}{RT}\right) (c_{H_2}^*) \exp\left(\frac{-0.5\Delta F_c}{RT}\right) \quad [49]$$

where ΔF_{ec} is the standard state free energy of activation for chemisorption and ΔF_c is the standard state free energy of chemisorption from the gas state.¹⁸ Thus the activation overvoltage at the anode can be defined as

$$\eta_{act anodic} = \frac{RT}{2F} \left\{ \ln \left[4FAk^0 \exp\left(\frac{-(\Delta F_{ec} - 0.5\Delta F_c)}{RT}\right) (c_{H_2}^*) \exp\left(\frac{-0.5\Delta F_c}{RT}\right) \right] - \ln(i) \right\} \quad [50]$$

The variation in hydrogen concentration in the inlet fuel feed for most low pressure fuel cell applications is expected to be relatively small ($x_{H_2} > 0.75$, 2 atm $< P < 4$ atm), and the resulting change in activation overvoltage will be less than 0.010 V. Thus for most empirical modeling applications the hydrogen concentration term in Eq. 50 can be approximated as a constant.

The total activation overvoltage for a fuel cell is the sum of the anodic and cathodic overvoltages, in parametric form

$$\eta_{act total} = \eta_{act anodic} + \eta_{act cathodic} = \xi_1 + \xi_2 T + \xi_3 T [\ln(c_{O_2}^*)] + \xi_4 T [\ln(i)] \quad [51]$$

where the constant parametric coefficients are

$$\xi_1 = \left(\frac{-\Delta F_e}{\alpha_c n F} \right) + \left(\frac{-\Delta F_{ec}}{2F} \right) \quad [52a]$$

$$\xi_2 = \frac{R}{\alpha_c n F} \ln [nFAk^0 c_{H^+}^{*1-\alpha_c} (c_{H_2O}^*)^{\alpha_c}] + \frac{R}{2F} [\ln(4FAk^0 c_{H_2}^*)] \quad [52b]$$

$$\xi_3 = \frac{R}{\alpha_c n F} (1 - \alpha_c) \quad [52c]$$

$$\xi_4 = -\left(\frac{R}{\alpha_c n F} + \frac{R}{2F} \right) \quad [52d]$$

Ohmic Resistance Losses

Ohmic polarization is expected to result from resistance to proton transfer in the solid polymer membrane and resistance to electron transfer in the graphite collector plates and the electrodes. Resistance to electron transfer is expected to be approximately constant over the operating range ($328 \text{ K} \leq T \leq 358 \text{ K}$), and the resultant voltage loss can be determined using an Ohm's law equation of the form

$$\eta_{\text{ohmic}}^{\text{electronic}} = -i \cdot R^{\text{electronic}} \quad [53]$$

where $R^{\text{electronic}}$ is the resistance to electron transfer.

Resistance to proton transfer through the membrane cannot be considered to be constant. The flux of protons through the membrane normal to the electrode faces can be described using the Nernst-Planck equation

$$N_i = -z_i \frac{F}{RT} D_i c_i \frac{\partial \Phi}{\partial z} - D_i \frac{\partial c_i}{\partial z} + c_i v \quad [54]$$

where Φ represents the electrical potential. The relationship between the flux of mobile ions and the current density can be defined as

$$I = F \sum_i z_i N_i \quad [55]$$

Substituting this equality into Eq. 54 and rearranging, one obtains

$$\frac{\partial \Phi^{\text{protons}}}{\partial z} = -\frac{I}{\kappa} - \frac{F}{\kappa} \left(\sum_i z_i D_i \frac{\partial c_i}{\partial z} \right) + \frac{F}{\kappa} \left(\sum_i z_i c_i \right) v \quad [56]$$

where κ is the membrane conductivity, defined as

$$\kappa = \frac{F^2}{RT} D_{\text{H}^+} c_{\text{H}^+} \quad [57]$$

All three components of Eq. 56 will be non-zero. An electrical potential gradient will exist to create migration, a proton concentration gradient will exist to create diffusion of protons from the anode to the cathode, and diffusion and absorption of water in the membrane will lead to a convective proton flux. Integration yields

$$\Delta \Phi^{\text{protons}} = \eta_{\text{ohmic}}^{\text{protons}} = \int_0^{\ell_m} \frac{RTI}{F^2 D_{\text{H}^+} c_{\text{H}^+}} dz - \int_0^{\ell_m} \frac{RT d c_{\text{H}^+}}{R c_{\text{H}^+} dz} dz + \int_0^{\ell_m} \frac{RT v}{F D_{\text{H}^+}} dz \quad [58]$$

where ℓ_m is the thickness of the membrane.

The total drop in potential due to ohmic resistance will result from the combined resistance to electron and proton transport and can be defined via an Ohm's law relation

$$\eta_{\text{ohmic}}^{\text{total}} = \eta_{\text{ohmic}}^{\text{electronic}} + \eta_{\text{ohmic}}^{\text{protons}} = -i \cdot R^{\text{internal}} \quad [59]$$

where R^{internal} is the total internal resistance of the fuel cell. Assuming that the current normal to the cathode face is uniform in the membrane, and using Eq. 58 and 59, one obtains

$$R^{\text{internal}} = R^{\text{elect.}} + \frac{RT}{F^2 A} \int_0^{\ell_m} \frac{dz}{D_{\text{H}^+} c_{\text{H}^+}} - \frac{RT}{F i} \int_0^{\ell_m} \frac{dc_{\text{H}^+}}{c_{\text{H}^+} dz} dz + \frac{RT}{F i} \int_0^{\ell_m} \frac{v dz}{D_{\text{H}^+}} \quad [60]$$

The diffusivity and concentration of protons, as well as the convective proton velocity, will all be dependent upon the water content and distribution in the membrane. While many papers dealing with issues of water distribution in the membrane have been presented,^{13,14,20,21} there is a very limited understanding of the interaction of migration, diffusion, and convection processes and their overall effect on water content and distribution. The complexity of water distribution arises from the interdependence of all of the parameters involved. With the experimental apparatus used in our lab, there is no means of determining the flux of water vapor into the membrane via absorption, or the level

of hydration across the membrane. Thus it is not possible to separate the various phenomena of water transport in the membrane so as to make an informed assessment of how the three proton transport processes interact. Instead, it is simply observed that the water content and distribution in the membrane will be in some way dependent upon the operating current and temperature. (Diffusivities and water partial pressures vary with temperature, while proton fluxes, and therefore water fluxes, will vary with the current). Equation 60 would thus indicate that the internal resistance of the fuel cell will also be in some way dependent upon the current and the temperature.

In the absence of an explicit mechanistic model, linear regression techniques are used to determine a best fit parametric model for the internal resistance. A model considering the main, interaction, and quadratic effects of current and temperature would have the form

$$R^{\text{internal}} = \gamma_1 + \gamma_2 T + \gamma_3 i + \gamma_4 T i + \gamma_5 T^2 + \gamma_6 i^2 \quad [61]$$

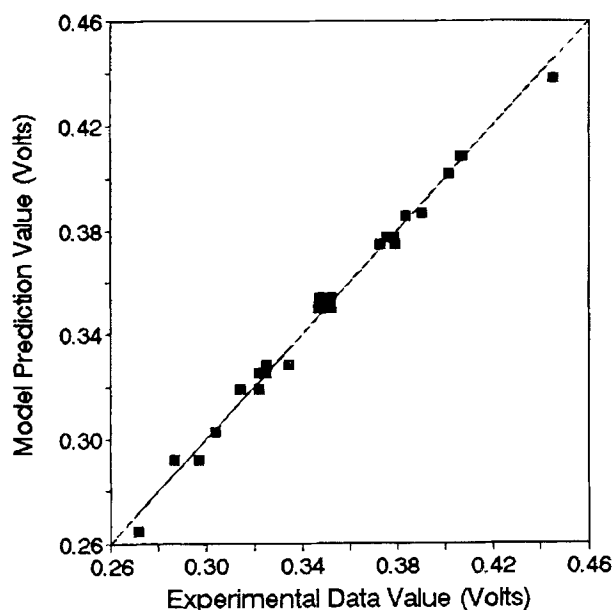


Fig. 1. Parity plot for activation overvoltage of experimental data as a function of model predictions for the Ballard Mark IV solid polymer fuel cell.

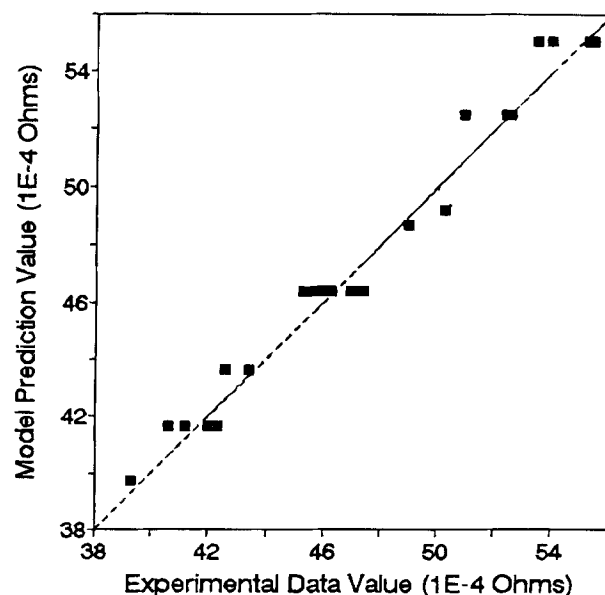


Fig. 2. Parity plot for fuel cell resistance of experimental data as a function of model predictions for the Ballard Mark IV solid polymer fuel cell.

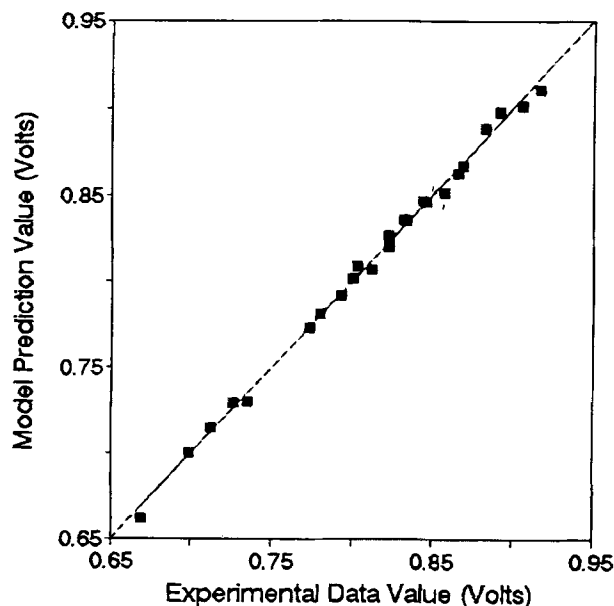


Fig. 3. Parity plot for total output voltage of experimental data as a function of model predictions for the Ballard Mark IV solid polymer fuel cell.

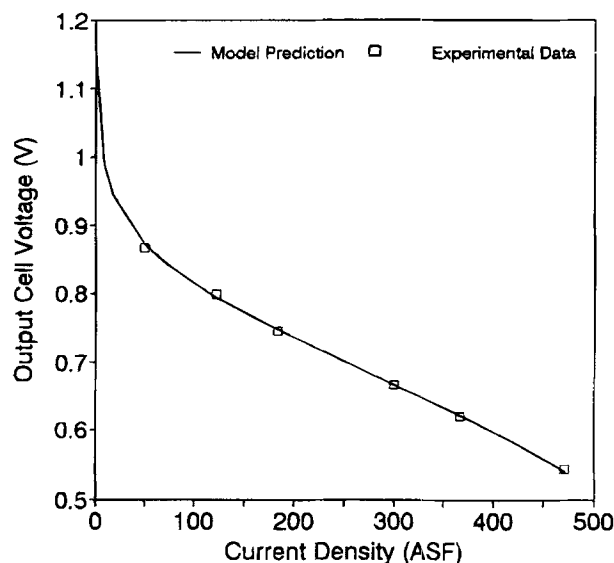


Fig. 4. Predicted and experimental response in output voltage as a function of operating current for the Ballard Mark IV solid polymer fuel cell.

where the γ values represent constant parametric coefficients. Equation 61 shows the most expansive form which a second-order parametric model would take if the resistance mapped linearly with current and temperature. Such a relationship will not necessarily apply, however, and it is possible that the relationship to either of these variables might be logarithmic, exponential, or otherwise.

Empirical Application of the Model

If all of the parameters in Eq. 37, 51, 52, 59, and 61 were known, they could be combined as shown in Eq. 38 to define fully the output voltage of the fuel cell. Unfortunately, not all of the parameters are directly available, and it is necessary to use empirical means to estimate values for these unknown terms. Details of this approach as well as the statistical evaluation of the resulting model are presented elsewhere.^{8,9}

The parametric equations developed from the empirical analysis were

$$\eta_{\text{act}} = -0.9514 + 0.00312T - 0.000187T [\ln(i)] + 7.4 \cdot 10^{-5}T [\ln(c_{\text{O}_2}^*)] \quad [62]$$

and

$$R^{\text{internal}} = 0.01605 - 3.5 \cdot 10^{-5}T + 8.0 \cdot 10^{-5}i \quad [63]$$

When coupled with Eq. 37 defining the thermodynamic equilibrium potential, these parametric models very accurately defined fuel cell performance over the experimental range ($328 \text{ K} \leq T \leq 358 \text{ K}$, $0.6 \text{ atm abs.} \leq p_{\text{O}_2}^{\text{inlet}} \leq 3.1 \text{ atm abs.}$, $30 \text{ ASF} \leq I \leq 450 \text{ ASF}$) as shown by the very good agreement between experimental and model-generated data displayed in Fig. 1 through 6. Figures 1 to 3 show parity plots of experimental data vs. model predictions for activation overvoltage, fuel cell resistance, and total output voltage. Figures 4 to 6 plot the predicted and experimentally determined response in output voltage for various independent changes in the operating current, temperature, and oxygen partial pressure.

Conclusions

The very good agreement between the empirical model and the experimental data indicates the value of developing a mechanistic basis for empirical modeling exercises. This approach is of considerable benefit in expanding the applicability and validity of the empirical model produced. In spite of the assumptions and approximations employed in the mechanistic approach, the predictions of the model seem to reflect the actual trends found in the experimental

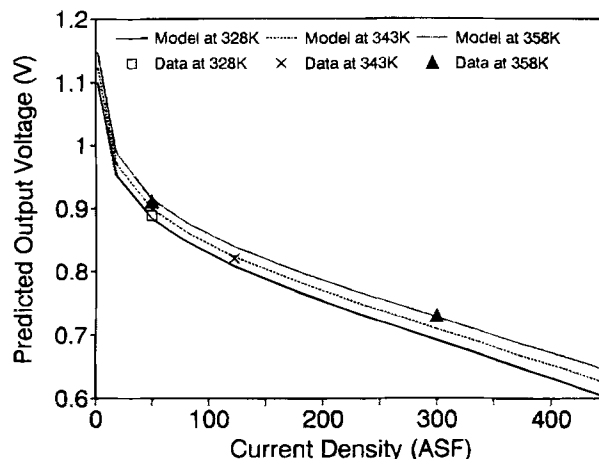


Fig. 5. Predicted and experimental response in output voltage as a function of temperature for the Ballard Mark IV solid polymer fuel cell.

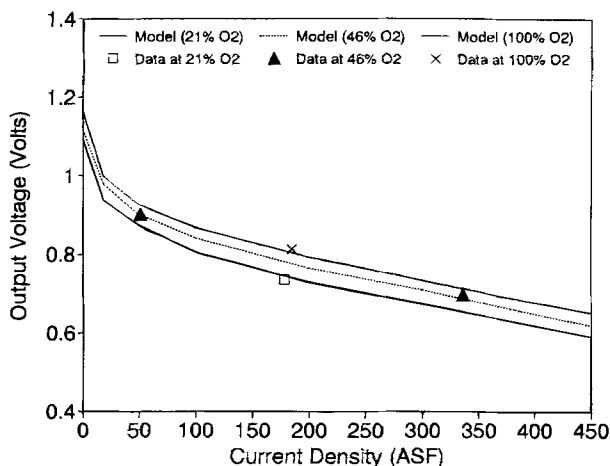


Fig. 6. Predicted and experimental response in output voltage as a function of the partial pressure of oxygen for the Ballard Mark IV solid polymer electrolyte fuel cell.

data. As a result of its mechanistic basis, the empirical model can be extrapolated beyond the experimental range with much more confidence than would otherwise be justified. Yet its application loses none of the ease and empirical validity for which empirical models are known. Thus it is concluded that the effort involved in coupling mechanistic and empirical approaches to model the performance of a PEM fuel cell is worthwhile, and yields a final model which will be both mechanistically valid and easily applied.

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LIST OF SYMBOLS

c_i	concentration of species i , mol/cm ³
D_i	diffusivity of species i , cm ² /s
D_{ij}^{eff}	effective binary diffusivity coefficient of the i - j gas pair, cm ² /s
E	thermodynamic equilibrium potential, V
E°	reference potential, V
E_0°	standard state reference potential, V
e^-	electrons
F	Faraday's constant, equal to 96,487 C/equiv.
H_i	Henry's law constant for species i , atm · cm ³ /mol
I	current density, A/cm ²
l_d	thickness of the gas diffuser, cm
l_t	thickness of the water film, cm
i	current, A
i_0	exchange current, A
K	proportionality constant
N_i	superficial gas flux of species i , mol/[(cm ²)(s)]
n	number of equivalents involved in a reaction
P	pressure, atm
p_i	partial pressure of component i , atm
R	gas constant, 8.3143 J/(mol)(K), or other appropriate units
R_{ext}	external electrical resistance, Ω
R_{int}	internal electrical resistance, Ω
ΔS_0°	standard state entropy change, kJ/mol
T	temperature, in Kelvin unless otherwise stated
t	time, s
x_i	mole fraction of species i
z	distance in the "horizontal direction" (direction of diffusion of gases), cm
z_i	charge on species i

Greek

α	transfer coefficient
ϵ	diffusivity correction factor = 0.3
η	overvoltage, V
η_{act}	overvoltage due to activation, V
η_{ohmic}	overvoltage due to ohmic resistance, V
μ	viscosity of water, g/(cm)(s)

Superscripts and Subscripts

act	activation
channel	conditions at the interface of the cathode gas flow channel and the cathode itself
CO ₂	carbon dioxide
crit	critical values
dry	dry gas conditions
eff	effective (i.e., average) conditions

equil. equilibrium

H₂ hydrogen

H₂O water

hum humidified gas conditions

in gas conditions at the inlet to the fuel cell

interface conditions at the interface of the gas diffusion cathode and the water film surrounding the catalyst sites

N₂ nitrogen

O₂ oxygen

out gas conditions at the outlet of the fuel cell

sat saturation conditions

* effective value

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